2026 9478 H2 Physics Preliminary Examination

Paper 3 (Mark Scheme)

1 **(a)** When the tension in the rope is more than the weight of the person, the resultant force acting on the person is opposite in direction to his velocity and the person is decelerating. B1

When the tension in the rope is less than the weight of the person, the resultant force acting on the person is in the same direction as his velocity and the person is accelerating. B1

Thus, the person will have maximum speed when the resultant force acting on him is zero and with a corresponding acceleration of zero.

(b) Man is moving at maximum speed when resultant force acting on him is zero.

$$mg = ke$$

$$e = \frac{mg}{k} = \frac{70.0 \times 9.81}{120} = 5.72 \text{ m}$$ C1

Applying Principle of Conservation of Energy,
Total energy before stepping off = Total energy at maximum speed

$$0 + 0 + 0 = \frac{1}{2} m v^2 + \frac{1}{2} k e^2 - mg(l + e)$$

$v = \sqrt{\square}$ C1

$\iquest \sqrt{\square}$

$= 23.4 \text{ m s}^{-1}$ A1

(c) Applying Principle of Conservation of Energy,
GPE at highest point = EPE at lowest point

$$mg(l + e) = \frac{1}{2} k e^2$$ M1

$$70 \times 9.81 \times (25 + e) = 0.5(120)(e^2)$$

$e = 23.6 \ m$ A1

2 **(a)** either: use conservation of energy ½ $mv^2 = m$ (9.81) (0.86)
or: use equation of motion $v^2 = 0^2 + 2$ (9.81) (0.86) M1

leading to 4.1 m s^{-1} A0

(b) 4.1 m s^{-1} B1

(c) velocity parallel to slope remains constant after collision M1

velocity perpendicular to slope retains the same magnitude after collision (since elastic) M1

so, velocity before and after collision is 30° to the normal of the slope A1

(d) time of flight = 0.6164 s (using equation of motion) C1

Horizontal distance between P and R = 2.189 m C1

QR = 2.189 – 1.2 cos 30° = 1.149 m = 1.1 m or 1.15 m A1

3 **(a)** **(i)** $f = \frac{1}{T} = \frac{1}{0.6} = 1.67$ Hz B1

(ii) maximum speed = $\omega x_0 = 2\pi(1.67) \times 0.010$ C1

Total energy
$= 0.5mv_0^{\,2} = 0.5m\omega^2 x_0^{\,2}$
$= (0.5)(0.34)(2\pi(1.67))^2 (0.010)^2$ M1

$= 0.00187$ J B1

(b) **(i)** Forced oscillations/Resonance B1

(ii) 1.67 Hz B1

(c) amplitude of oscillations at all frequencies decreases (lines do not intersect) B1

peak oscillation at frequency < original B1

4 **(a)** figure shows an interference pattern (which can only occur when waves from both sources are coherent). B1

(b) intensity at P or $x = 0$ is a minimum B1

since point is <u>equidistant from speakers so path difference = 0</u> B1

intensity minimum if and only if <u>waves from speaks are emitted anti-phase</u> B1

(c) $x = \lambda D / a$

$0.040 = \lambda\ (1.5) / 0.15\ (\Longrightarrow \lambda = 4.0 \times 10^{-3}$ m) M1

$v = f \lambda$

$f = 340 / 4.0 \times 10^{-3} = 85\ 000$ Hz A1

(d) intensity graph is only zero at the 12 cm mark which is farther from L_1 than L_2. M1

Since intensity decreases with distance, speaker L_1 has a higher power. A1

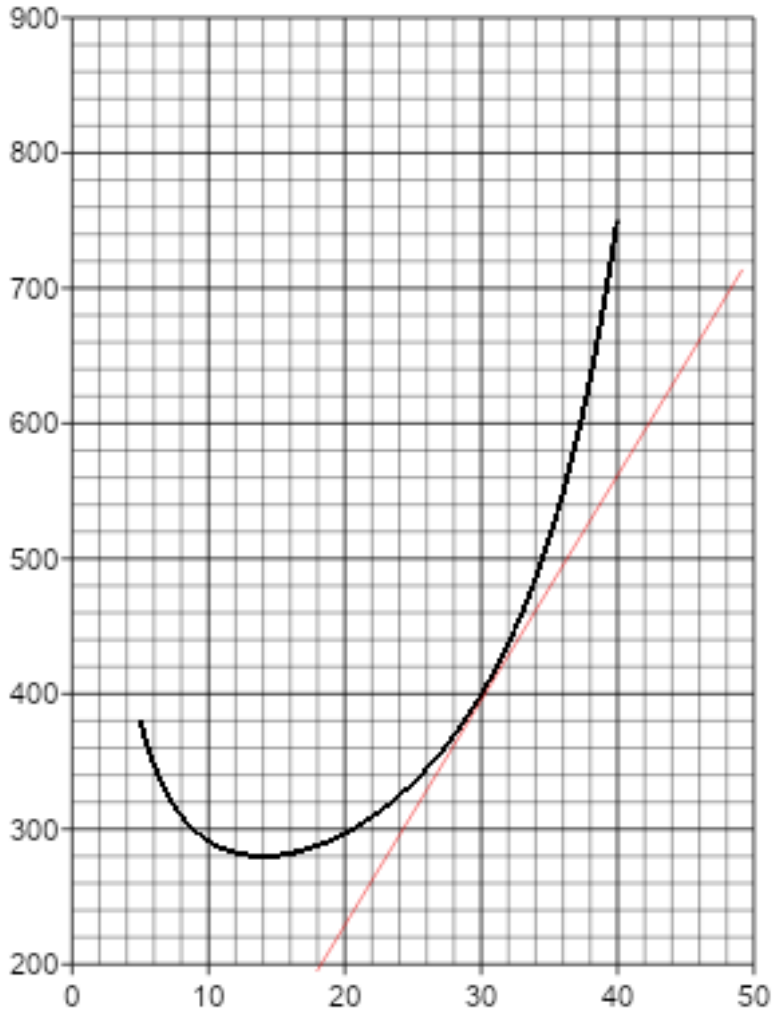
5 **(a)** The potential of both charge near their surface is positive B1

(b) From x = 5 cm to about 14 cm, (resultant) electric field (strength) is towards B and decreasing in magnitude. B1

(Resultant) electric field (strength) is zero at *x* ~14 cm B1

From x ≈ 14 cm to x = 40 cm, (resultant) electric field (strength) is towards A and increasing in magnitude. B1

(c) show construction and correct calculation using $\Delta V / \Delta x$ M1
(0.21,250), (0.44,630)
E = 1650 N C^{-1}



(d) Consider x = 0.30 m, E = 1650 NC^{-1} (or any other appropriate point)

$$\frac{-1}{4\pi\varepsilon_0}\left(\frac{Q_A}{0.30^2}\right)+\frac{1}{4\pi\varepsilon_0}\left(\frac{8.0\times10^{-9}}{(0.50-0.30)^2}\right)=1650$$ M1

Q_A = ¿ 1.48 nC or 1.5 nC A1

(e) Minimum kinetic energy required to reach A
= gain in electric potential energy from surface of A to neutral point
= e ($V_{\text{neutral point}} - V_{\text{surface of B}}$)

Change in potential = (280 – 750) C1

From energy conservation / loss in k.e. = gain in e.p.e. M1

$$\frac{1}{2}\left(9.11 \times 10^{-31}\right) v^2 = e\left(750 - 280\right) = 7.52 \times 10^{-17}$$

v = 1.28 × 10^7 m s^{-1} or 1.3 × 10^7 m s^{-1} A1

6 **(a)** Since reading decreases, magnetic force is upwards. B1

From Fleming's left hand rule, magnetic field points/directed into the page. B1

(b) magnetic force on coil F ¿ $\frac{(125.76 - 101.35) \times 10^{-3}}{2} \times 9.81 = 0.11973$ N C1

$F = NBIL$ M1

$B = \frac{0.11973}{60 \times 6.0 \times 0.085}$

$B = 3.91 \times 10^{-3}$ T A1

(c) forces on PS & RQ act horizontally / act in opposite directions / cancels off B1

no effect on the reading B1

7 **(a)** **(i)** max flux linkage = $64 \times 0.080 \times 0.71 \times 10^{-4}$ M1

$= 3.6 \times 10^{-4}$ Wb A1

(ii) *E* is zero from time 0 to time *t*

E has constant non-zero value between *t* and 2*t*, B1
E opposite sign between 2*t* and 4*t* B1

(b) decrease in flux density of solenoid (towards end) results in decrease in flux linkage in coil B1

According to Faradays' Law, induced emf in coil leads to (induced) current because Coil C is connected B1

which causes magnetic field around coil that interact with solenoid magnetic field to create attractive force B1

8 **(a)** **(i)** *collision rate:* molecule takes more time to complete one round trip and return to hit same wall B1

leads to decrease in collision rate B1

gas pressure: (from eq. 8.2) smaller force due to lower collision rate (and change in momentum per collision remains constant) B1

so pressure (= force / area) decreases B1

(ii) *collision rate:* (effective free space inside the container is reduced when volume of the gas molecules is taken into account,)

molecule travels less distance to complete one round trip and return to hit same wall B1

leads to increase in collision rate B1

gas pressure: (from eq. 8.2) larger force due to higher collision rate (and change in momentum per collision remains constant) B1

so pressure (= force / area) increases B1

(b) **(i)** calculate product of *p* and *V* of any two points e.g. (1.0, 3.2), (1.44, 2.0), (1.5, 1.9), (1.8, 1.5), (1.9, 1.4) M1

valid conclusion e.g. since product of *p* and *V* are not equal, temperature of process is not constant A1

(ii) average or mean kinetic energy (not internal energy) proportional to thermodynamic temperature B1

gas molecules transfer momentum to the piston during collision / molecules rebound from the piston with smaller momentum M1

average or mean (translational) kinetic energy of the gas molecules decreases A1

so, temperature decreases A0

(iii) **1.** substitute in pV^{β} any two points from graph [e.g. (1.0, 3.2), (1.44, 2.0), (1.5, 1.9), (1.8, 1.5), (1.9, 1.4)] and equate C1

correct manipulation to obtain ¿ 1.29 or 1.3 A1

$p_1 V_1^{\beta} = p_2 V_2^{\beta}$ $\quad \frac{V_1^{\beta}}{V_2^{\beta}} = \frac{p_2}{p_1}$ $\quad \left(\frac{V_1}{V_2}\right)^{\beta} = \left(\frac{p_2}{p_1}\right)$

$\beta \ln\ln\left(\frac{V_1}{V_2}\right) = \ln\ln\left(\frac{p_2}{p_1}\right)$

$$\beta = \frac{\ln\ln\left(\frac{p_2}{p_1}\right)}{\ln\ln\left(\frac{V_1}{V_2}\right)} = \frac{\ln\ln\left(\frac{1.4}{3.2}\right)}{\ln\ln\left(\frac{1.0}{1.9}\right)}$$

2. no unit / dimensionless B1

(iv) calculate area under graph to obtain work done on gas between 1700 J and 1800 J A2
(if value 1.6 – 1.7 kJ OR 1.8 – 1.9 kJ, award one mark only)

Zero mark if values outside range.
Deduct one mark if answer includes negative sign.

V / 10^{-3} m^3	1.0	1.22	1.28	1.4	1.64	1.9
p / MPa	3.2	2.3	2.1	1.8	1.5	1.4
area or trapezium / 10^3 J	–	0.605	0.132	0.234	0.396	0.377

work done on gas = total area = 1.74 kJ
($|W_B| < |W_A|$. For comparison, work done by process A = 1.89×10^3 J.)

(v) [Thinking process:

- Apply first law to the cyclical process

process	**increase internal energy**	**heat supplied**	**work done on gas**
A	negative	0	negative
B	positive	positive	positive
net change of quantity	0	positive	negative (area of A > B)

- net work done by gas is positive
- heat supplied to gas in process B

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net work done by gas = heat supplied (during process B) M1

resulting in 100% efficiency, hence not possible A1

9 **(a)** **(i)** $E = hf = hc / \lambda$, so $E\lambda = hc$ with correct substitution of h and c M1

show conversion of E to eV and λ to nm correctly C1

$E\lambda = (6.63 \times 10^{-34})\ (3.00 \times 10^{8})\ (10^{9}) / 1.60 \times 10^{-19}$

$E\lambda = 1243$ eV nm A0

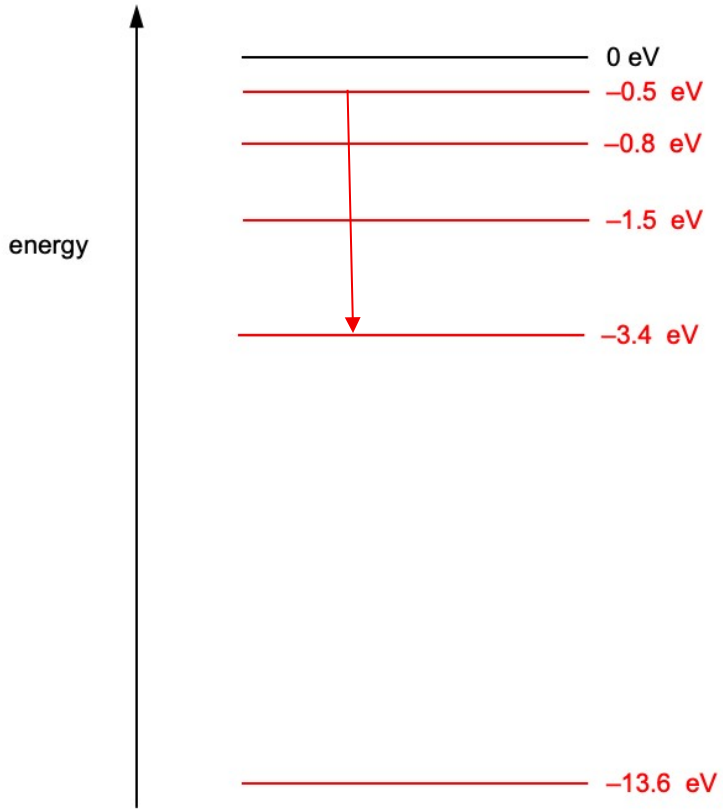
(ii) photon energy of 95.0 nm = 1243 / 95.0 = 13.1 eV A1

photon energy of 97.3 nm = 1243 / 97.3 = 12.8 eV A1

(iii) lowest energy at –13.6 eV B1

four energy levels (–0.5 eV, –0.8 eV, –1.5 eV and –3.4 eV) between 0 and –13.6 eV, deduct 1 mark for each missing level B3

increasing spacing as energy levels becomes lower B1



(iv) $E = 1243 / 430 = 2.89$ eV
arrow from –0.5 eV to –3.4 eV B1

(b) **(i)** wavelength of $\psi_B < \psi_A$ B1

since energy proportional to (momentum)^2 and momentum inversely proportional to wavelength (of electron) B1

so ψ_B has higher energy B1

(ii) electron transit from $n = 3$ (ψ_B) to $n = 1$ (ψ_A) C1

photon energy $\Delta E_3 - E_1 = \frac{h^2}{8mL^2}\left(n_3^2 - n_1^2\right) = \frac{\left(6.63 \times 10^{-34}\right)^2}{8\left(9.11 \times 10^{-31}\right)\left(1.0 \times 10^{-10}\right)^2}\left(3^2 - 1^2\right)$ C1

photon energy = 4.8 (or 4.82, 4.83) × 10^{-17} J A1

(c) **(i)** $\Delta p_y = h / \Delta y = 6.63 \times 10^{-34} / 1.5 \times 10^{-10}$ M1

$\Delta p_y = 4.4 \times 10^{-24}$ N s or 4.42×10^{-24} Ns A1

(ii) *y*-momentum uncertainty gives electrons a momentum (or velocity) perpendicular to the original direction (or stte) M1

(random process, so beam spreads out) with some electrons going to +*y* and some to –*y* (or stte) A1